Experiment 5

Outliers Detection.

Code 1

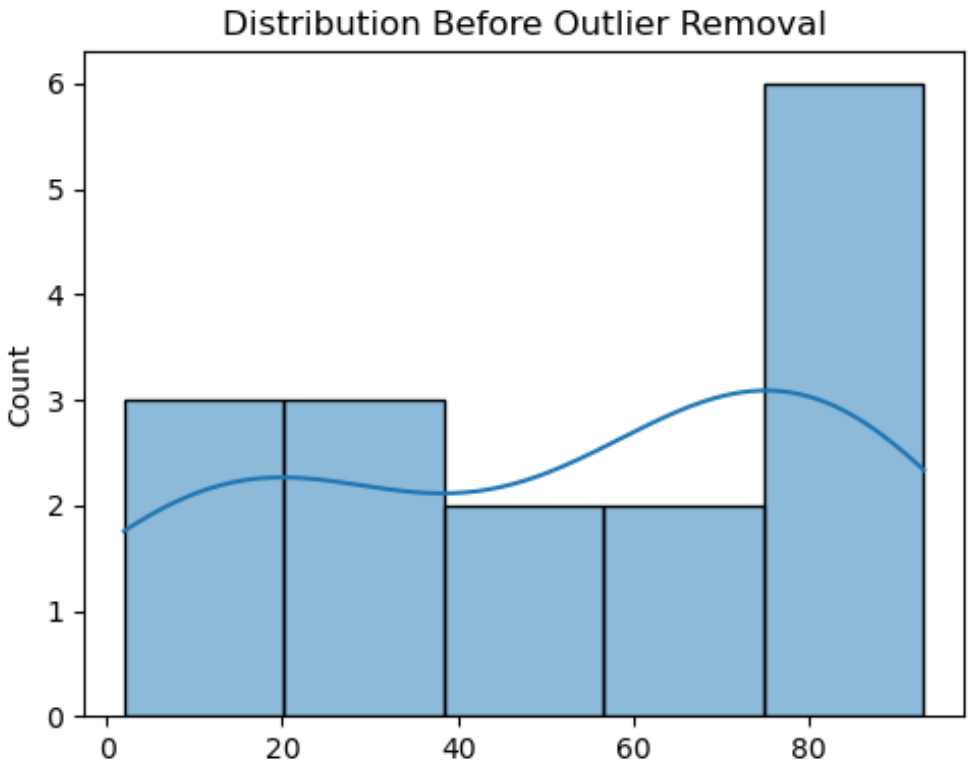
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
np.random.seed(42)

array = np.random.randint(1, 100, 16)
print("Generated array:")
print(array)
print()
print("Mean:", array.mean())
print()

print("25th percentile:", np.percentile(array, 25))
print("50th percentile:", np.percentile(array, 50))
print("75th percentile:", np.percentile(array, 75))
print("100th percentile:", np.percentile(array, 100))
print()

def outDetection(array):
 Q1, Q3 = np.percentile(array, [25, 75])
 IQR = Q3 - Q1
 lr = Q1 - (1.5 * IQR)
 ur = Q3 + (1.5 * IQR)
 return lr, ur
lr, ur = outDetection(array)
print("Lower Range, Upper Range:", (lr, ur))
print()

plt.figure(figsize=(5, 4))
sns.histplot(array, kde=True)
plt.title("Distribution Before Outlier Removal")
plt.tight_layout()
plt.show()

new_array = array[(array > lr) & (array < ur)]
print("Array after removing outliers:")
print(new_array)
print()

plt.figure(figsize=(5, 4))
sns.histplot(new_array, kde=True)
plt.title("Distribution After Outlier Removal")
plt.tight_layout()
plt.show()
print("Number of outliers removed:", len(array) - len(new_array))
```

Output

```
Generated array:
[52 93 15 72 61 21 83 87 75 75 88 24  3 22 53  2]

Mean: 51.625

25th percentile: 21.75
50th percentile: 57.0
75th percentile: 77.0
100th percentile: 93.0

Lower Range, Upper Range: (np.float64(-61.125), np.float64(159.875))
```

Output

```
<Figure size 500x400 with 1 Axes>
```

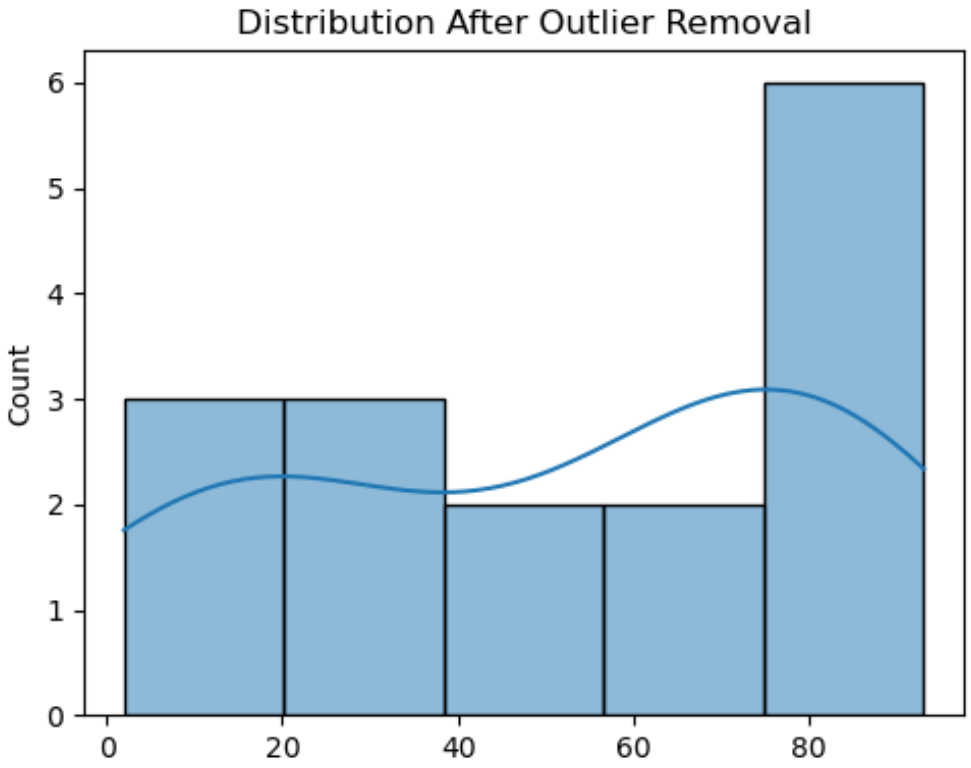


Output

```
Array after removing outliers:
[52 93 15 72 61 21 83 87 75 75 88 24  3 22 53  2]
```

Output

```
<Figure size 500x400 with 1 Axes>
```



Output

```
Number of outliers removed: 0
```